Lars Larsen

Embedded C++ & FPGA Engineer — Real-Time Video

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SUMMARY

Embedded engineer with eight years of C/C++ firmware on Zynq-7000 (bare-metal and FreeRTOS), STM32F4 and RP2040/RP2350, and the FPGA video systems beside it. Primary author of LZX's shared embedded framework and its drivers across TBC2, Memory Palace, Chromagnon and Videomancer. Verifies with simulation, bit-exact models and hardware-in-the-loop rigs.

EXPERIENCE

Founder & Principal Engineer, Firmware / FPGA — LZX Industries

Portland, OR · 2011 – Present

Video synthesizer and video-processing instrument maker (DBA 2011, LLC 2017). Lead designer of every product in a three-person core team (partners on firmware/HDL and board layout).

- Led 100+ hardware products from concept to production across six Eurorack series, standalone instruments, DIY kits, cases and power: ~240 production circuit-board designs and ~140 metal front panels; $6.4M direct-to-consumer sales (2018–2026) plus a dealer channel.
- Videomancer (shipping since late 2025): architected a standalone FPGA video synthesizer — iCE40 HX4K + RP2040; ADV7611/ADV7513 HDMI in/out; ADV7181C + ADA4412-3 analog input; ADV7393 SD/ED/HD encoder; composite, S-Video, YPbPr and RGB — with user-loadable effects in Ed25519-signed packages, 69 public release candidates, and an open GPL-3.0 SDK.
- Embedded C++ since 2018: primary author of firmware for TBC2 and Memory Palace (Zynq-7010, bare-metal and FreeRTOS), Scrolls (STM32F411), and Chromagnon and Videomancer (RP2040); evolved a shared application engine (state machines, UI, presets, modulation, MIDI, USB) through three generations into a C++20 framework with no exceptions, RTTI, STL containers or heap allocation.
- ESG3 Encoder & Sync Generator (Gen3, 2022): pipelined the sync timing path on an iCE40 Ultra so SD/HD and tri-level sync met timing at HD pixel clocks with fixed, phase-accurate latency; VCXO PLL genlock. The FPGA12 iCE40 daughterboard is shared across Gen3 FPGA modules and power/sync entry products.
- Bare-metal drivers for video decoders, encoders and HDMI transceivers (ADV7181C, ADV7393, ADV7611, ADV7513, THS7353, ADA4412-3), DACs/ADCs (MCP4728, ADS1115, MCP3208), I/O expanders and an audio codec (TLV320AIC3104), with host-side mock-based unit tests and PC simulators.
- Chromagnon: re-architected from a Zynq SoM to iCE40HX4K + RP2040. iCE40 RTL recovers timing from embedded SAV/EAV codes, converts YCbCr to RGB and genlocks 15 SD/ED/HD formats to 74.25 MHz; brought up passthrough, Sync Out and feedback across all 15 timings.
- Chromagnon (2026): Python instrument model bit-exact to the iCE40 RTL and GHDL, with nine block test runners (76/76 injected faults caught in simulation), Rigol MSO5000 automation over SCPI, and 88 golden-vector tests for its WebGL browser port.
- Software: led the Shopify Hydrogen/React storefront since 2023 (~58k lines, Playwright); in 2026, with AI coding agents, a Python operations CLI (~1,000 commands, ~3,500 tests) integrating Shopify, InvenTree, ERPNext, JLCPCB and UPS, a Rust/Tauri firmware updater, and self-hosted Forgejo CI and Authentik SSO.

Operations Manager — Darkplace Manufacturing

Portland, OR · 2015 – 2017

Manufacturing arm for the LZX Industries product line.

- Managed operations, purchasing and customer communication for a manufacturing team of 10+ people building the LZX product line, while continuing as its product designer.

Earlier

- Multimedia Programmer, University of North Texas, Denton TX — 2008–2010, 2011–2015
- Lead Service Technician, Switched On (synthesizer sales & repair), Austin TX — 2010

- New Media Developer, FUNimation Entertainment, Flower Mound TX — 2007–2008
- Multimedia Developer, TargetScope, Plano TX — 2006–2007

SELECTED PROJECTS

- **Videomancer SDK (GPL-3.0)** — iCE40 video platform, program ABI and signed package format, GHDL image tester; ~190 C++ and ~350 VHDL unit tests. github.com/lzxindustries/videomancer-sdk
- **Chromagnon web simulator** — In-browser WebGL port of the Chromagnon instrument model, checked against golden vectors. lzxindustries.net/instruments/chromagnon/tools/simulator
- **Videomancer community programs** — Open VHDL video effects by outside contributors, built on the SDK's toolchain. github.com/lzxindustries/videomancer-community-programs

SKILLS

Firmware & FPGA	C/C++ (C++17/20, freestanding, no-heap), Zynq-7000 bare-metal and FreeRTOS, STM32F4, RP2040/RP2350, AVR, TinyUSB, Ed25519 signing; VHDL, Verilog, GHDL, VUnit, Yosys/nextpnr (iCE40; ECP5 designs), Vivado (Zynq-7000)
Electronics	Analog/mixed-signal and video (SD–1080p, composite/S-Video/YPbPr/RGB/HDMI, genlock), discrete high-speed analog, power distribution, ngspice/SPICE and IBIS, DipTrace, Eagle, KiCad + own fork, DFM, BOM cost engineering, bring-up, oscilloscope/SCPI automation
Software	Python (NumPy, OpenCV, PySide6), TypeScript/React (Shopify Hydrogen/Remix, Vitest, Playwright), Docker, Forgejo CI, Linux servers, AI-assisted development (Claude Code, Cursor, MCP)
Product & operations	Concept-to-production ownership, PLM (MRPeasy, InvenTree), ERP, supplier and contract-manufacturer management, pricing, documentation, open-source hardware programs
Mechanical	Inventor, FreeCAD, build123d, sheet-metal enclosure design, injection-molded knobs, front-panel graphics, Blender product rendering

EDUCATION

- B.A., Radio, Television & Film — University of North Texas, 2006